ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

General Chemestry I

Course notes

Atomic structure · molecules · Bonds

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1 Atomic Structure

An atom is composed of a **nucleus** (protons + neutrons) surrounded by an **electron cloud**. The **atomic number** Z (= no. protons = no. electrons for a neutral atom) uniquely identifies the element; the **mass number** A = protons + neutrons. The notation is ${}^{A}_{Z}X$.

Particle	Charge	Mass (kg)	Location
Proton	$+e$	$1{,}673 \times 10^{-27}$	Nucleus
Neutron	0	$1{,}675 \times 10^{-27}$	Nucleus
Electron	$-e$	$9{,}109 \times 10^{-31}$	Electron cloud

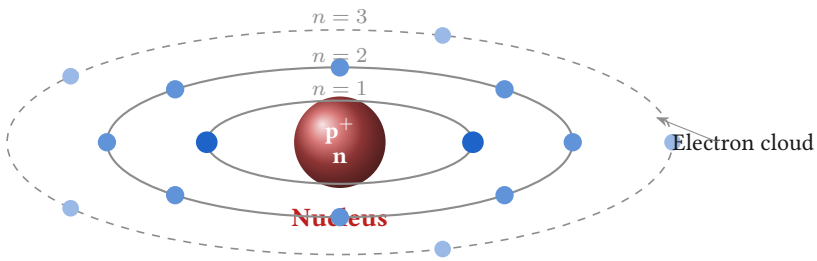

Figure 1: Structure of the atom: nucleus (protons p^+ and neutrons n) surrounded by electron shells $n = 1, 2, 3, \ldots$

1.1 Isotopes and Atomic Mass

Isotopes are atoms of the same element (same Z) with different numbers of neutrons ($n = A - Z$). The *relative atomic mass* is the weighted average of isotope masses over their natural abundances:

$$M = \sum_i f_i \cdot m_i$$

where f_i is the fractional abundance and m_i the mass of isotope i.

$M(\text{Cl}) = 0{,}7577 \times 34{,}969 + 0{,}2423 \times 36{,}966 \approx 35{,}45$ u

1.2 Mole and Avogadro's Number

The *mole* (mol) contains exactly $N_A = 6{,}022 \times 10^{23}$ entities (atoms, molecules, ions…). The *molar mass* M [g/mol] is numerically equal to the atomic mass in u.

$$n = \frac{m}{M} \qquad N = n \cdot N_A$$

The mole is the fundamental SI unit of amount of substance.

1.3 Fundamental Laws of Chemistry

Law of Conservation of Mass (Lavoisier): in a chemical reaction, the total mass of the reactants is exactly equal to the total mass of the products. Matter is neither created nor destroyed.

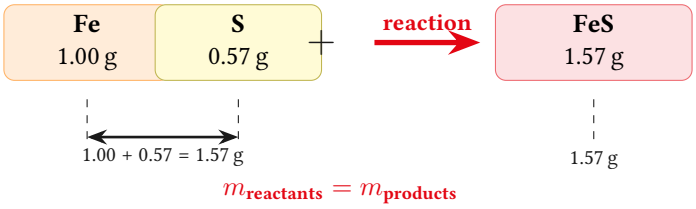

Figure 2: Example of Lavoisier's law: 1 g of iron + 0,57 g of sulfur → 1,57 g of iron sulfide.

Law of Definite Proportions (Proust): a chemical compound is always formed by the same elements combined in constant and definite mass ratios.

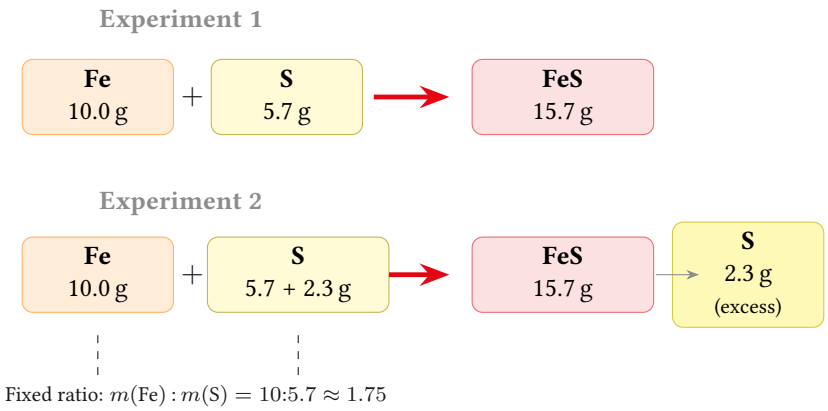

Figure 3: Proust's law: 10 g of iron react with 5,7 g of sulfur; the excess sulfur (2,3 g) does not react.

2 Electronic Configuration

Heisenberg's Uncertainty Principle: it is impossible to simultaneously know with arbitrary precision both the *position* and the *momentum* of an electron:

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$$

where $\hbar = h/2\pi$ is the reduced Planck constant. There are no well-defined orbits: instead, *orbitals* are used — regions of space in which the probability of finding the electron is $\geq 90\%$.

2.1 Quantum Numbers and Orbitals

An *atomic orbital* is a wave function ψ_{n,l,m_l}. Each electron is uniquely described by four quantum numbers:

- $n = 1, 2, 3, \ldots$ **(principal)**: determines the energy level and average distance from the nucleus.
- $l = 0, \ldots, n-1$ **(azimuthal)**: defines the shape of the orbital (s, p, d, f).
- $m_l = -l, \ldots, +l$ **(magnetic)**: specifies the spatial orientation.
- $m_s = \pm\frac{1}{2}$ **(spin)**: indicates the intrinsic rotation of the electron.

Type	l	No. orbitals	Max e^-	Shape
s	0	1	2	Spherical
p	1	3	6	Bilobal (3 orientations: x, y, z)
d	2	5	10	Four-lobed
f	3	7	14	Complex

The maximum number of electrons per shell n is $2n^2$.

2.2 Filling Rules

Aufbau Principle: electrons occupy orbitals in order of increasing energy (Klechkowski rule):

$$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < \cdots$$

Pauli Exclusion Principle: no orbital can hold more than two electrons; if two are present, they must have opposite spins.

Hund's Rule: among degenerate orbitals, electrons occupy distinct orbitals first with parallel spin, minimizing electrostatic repulsion.

2.3 Notation and Ionic Configuration

The configuration is written by listing subshells with a superscript for the number of electrons; the preceding noble gas is abbreviated in square brackets. For sulfur ($Z = 16$):

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4 \ \equiv\ [\text{Ne}]\ 3s^2\ 3p^4$$

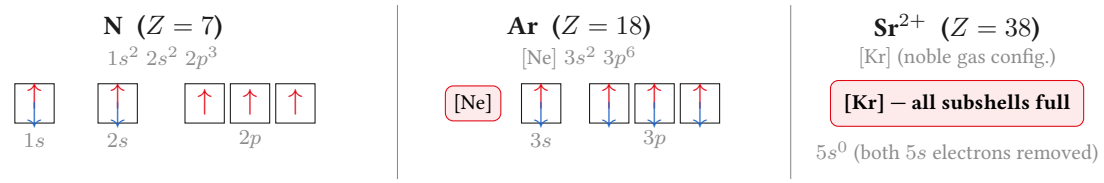

Figure 4: Filling diagrams for N ($1s^2\ 2s^2\ 2p^3$, Hund's rule), Ar (complete noble-gas configuration) and Sr^{2+} (isoelectronic with Kr after loss of both $5s$ electrons).

Cation: the atom *loses* electrons ⇒ positive charge. **Anion**: the atom *gains* electrons ⇒ negative charge. Electrons are always removed from or added to the highest-energy subshell.

Fe (Z=26) : [Ar] $3d^6\ 4s^2$ → Fe^{2+} : [Ar] $3d^6$ → Fe^{3+} : [Ar] $3d^5$.
In transition metals, $4s$ electrons are removed *before* $3d$ electrons.

3 Periodic Table and Periodic Properties

Elements are arranged in order of increasing Z. **Columns** (18 groups) collect elements with the same valence configuration; **rows** (7 periods) group elements with the same number of occupied shells.

3.1 Effective Nuclear Charge

The *effective nuclear charge* $Z_{\text{eff}} = Z - \sigma$ is the charge perceived by the valence electron after **shielding** σ by inner electrons. Z_{eff} **increases** across a period and remains **nearly constant** down a group.

3.2 Periodic Trends

Property	Across period (→)	Down group (↓)
Atomic radius	Decreases	Increases
Ionization energy	Increases	Decreases
Electron affinity	Increases	Decreases
Electronegativity (Pauling scale)	Increases (max F = 4,0)	Decreases (min Cs = 0,79)
Metallic character	Decreases	Increases

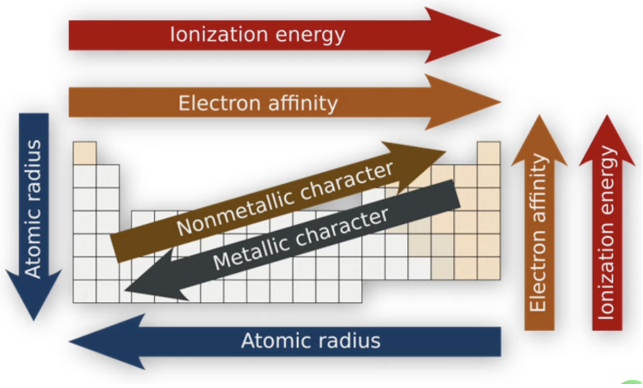

Figure 5: Summary of the main periodic trends.

4 Molecular Geometry and Polarity

4.1 VSEPR Theory

The **VSEPR** theory (*Valence-Shell Electron-Pair Repulsion*) states that electron pairs around a central atom — both bonding and lone pairs — arrange themselves in space to **minimise mutual repulsion**, maximising the angular distance between them.

The notation A X_n E_m describes the central atom A with n bonded atoms X and m lone pairs E. Lone pairs occupy more angular space than bonding pairs and **compress** bond angles.

AXE	El. gr.	Molecular geometry	Angle	Example
AX_2E_0	2	Linear	180	BeH_2, CO_2
AX_3E_0	3	Trigonal planar	120	BH_3, BF_3
AX_4E_0	4	Tetrahedral	109,5	CH_4
AX_3E_1	4	Trigonal pyramidal	107,3	NH_3
AX_2E_2	4	Bent	105	H_2O
AX_5E_0	5	Trigonal bipyramidal	90/120	PCl_5
AX_6E_0	6	Octahedral	90	SF_6

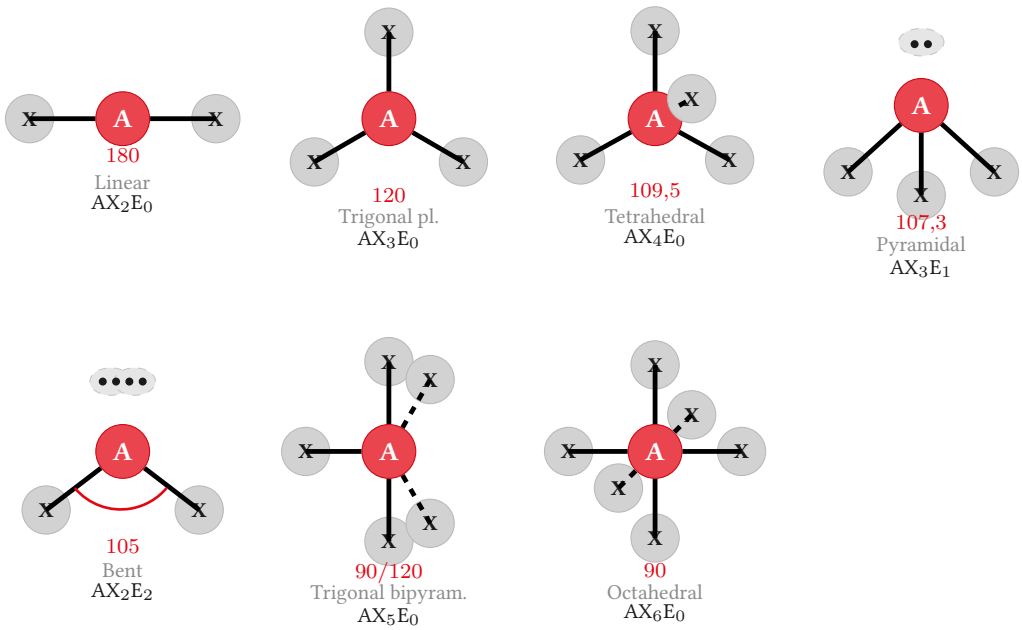

Figure 6: Main molecular geometries according to VSEPR theory. Lone pairs (E) are shown as dashed ellipses and compress bond angles.

4.2 Polarity of Molecules

A molecule is **polar** when it possesses a **permanent non-zero dipole moment** ($\vec{\mu}_{\text{tot}} \neq 0$): there is a persistent net separation of positive (δ^+) and negative (δ^-) charge. Two conditions must be satisfied *simultaneously*:

1. **Polar bonds**: the bonded atoms have a significant ΔEN, generating partial bond dipoles ($\vec{\mu}_i$).
2. **Asymmetric geometry**: the vector sum of the individual $\vec{\mu}_i$ does not cancel, yielding $\vec{\mu}_{\text{tot}} \neq \vec{0}$.

If the geometry is symmetric, even polar bonds can give $\vec{\mu}_{\text{tot}} = 0$ (**nonpolar** molecule).

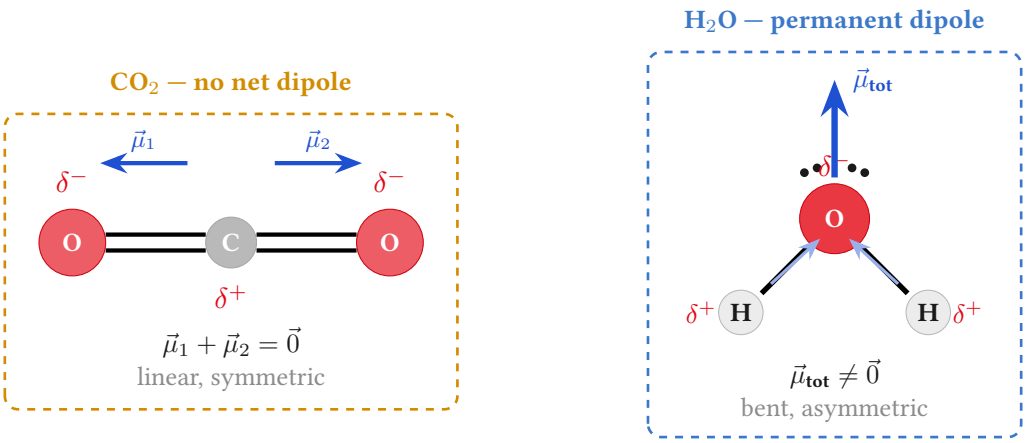

Figure 7: Comparison between a nonpolar molecule (CO_2, linear) and a polar molecule (H_2O, bent). In CO_2 the two dipoles cancel by symmetry; in H_2O the bent geometry produces a net dipole directed towards oxygen (δ^-).

Nonpolar ($\vec{\mu}_{\text{tot}} = 0$)	**Polar** ($\vec{\mu}_{\text{tot}} \neq 0$)
Polar bonds present, but the vector sum cancels by symmetry.	Asymmetric geometry: the resultant of the bond dipoles is non-zero.
E.g. CO_2 (linear): the two $\vec{\mu}$(C=O) oppose each other.	*E.g.* H_2O (bent): the two $\vec{\mu}$(O–H) do not cancel.
E.g. CCl_4 (symmetric tetrahedral): $\vec{\mu}_{\text{tot}} = 0$.	*E.g.* NH_3 (pyramidal), HCl (heteronuclear diatomic), HF.

A nonpolar bond ($\Delta EN \approx 0$) does not contribute to the dipole moment. Molecular polarity requires *both* polar bonds *and* asymmetric geometry: polar bonds alone are insufficient if the geometry is symmetric (as in CCl_4). Polarity also determines **solubility**: *like dissolves like* — polar solvents dissolve polar or ionic substances; nonpolar solvents dissolve nonpolar substances.

5 Lewis Structure

A Lewis structure is a graphical representation showing the distribution of **valence electrons** in a molecule or polyatomic ion. Bonds are indicated by dashes (each dash = 2 shared electrons), lone pairs by pairs of dots.

A **Lewis structure** is a representation that makes explicit:

- all **covalent bonds** between atoms (single, double or triple);
- all **lone pairs** (not involved in bonding);
- the **formal charge** (FC) of each atom:

$$\text{FC} = (\text{valence electrons}) - (\text{lone electrons}) - \tfrac{1}{2}(\text{bonding electrons})$$

The most stable structure minimises formal charges (ideally FC $= 0$) and, if residual, assigns negative charges to the most electronegative atoms.

Atoms tend to satisfy the **octet rule** (ns^2np^6, 8 valence electrons); H and He follow the **duet rule** ($1s^2$, 2 electrons).

Step-by-step procedure:

1. **Count** the total valence electrons.
2. **Build the skeleton**: least electronegative atom is central; H and F always peripheral.
3. **Complete the octets** of peripheral atoms.
4. **Place remaining electrons** on the central atom as lone pairs.
5. **Check the central octet**: if incomplete, convert lone pairs into multiple bonds.
6. **Calculate formal charges** and select the most stable structure.

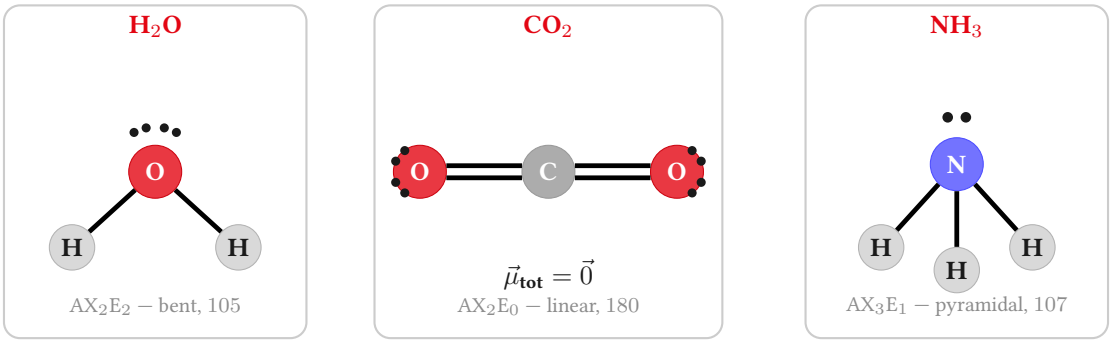

Figure 8: Lewis structures for H_2O, CO_2 and NH_3. Dots indicate lone pairs.

Exceptions to the octet rule.

- *Incomplete octet*: BF_3 (B has only 6 e^-), BeH_2 (Be has 4 e^-). These atoms are electrophilic.
- *Expanded octet*: SF_6, PCl_5, XeF_4. Third-period elements use vacant $3d$ orbitals.
- *Radicals*: NO, NO_2. Odd number of e^-, paramagnetic.

Resonance structures. When multiple equivalent Lewis structures describe the same molecule, the real molecule is a *resonance hybrid* with bonds of intermediate length and energy. Example: CO_3^{2-} has three equivalent resonance structures; C–O bond order $= 4/3 \approx 1{,}33$.

H_2O: valence $e^- = 2\times1 + 6 = 8$. Geometry $AX_2E_2 \Rightarrow$ **bent**, 105.
CO_2: valence $e^- = 4 + 2\times6 = 16$. Two double bonds. Geometry **linear**, 180.
$AsCl_3$: valence $e^- = 5 + 3\times7 = 26$. Geometry $AX_3E_1 \Rightarrow$ **trigonal pyramidal**.

6 Chemical Bonds

A *chemical bond* is an electrical interaction that holds atoms together in a molecule or solid. The type of bond depends mainly on the **electronegativity difference** ΔEN between the atoms involved:

- **Intramolecular bonds**: hold atoms together *within* a molecule. Typical energies: 150–1000 kJ mol^{-1}.
- **Intermolecular bonds**: act *between* distinct molecules. Typical energies: 0,1–40 kJ mol^{-1}.

6.1 Intramolecular Bonds

Type	ΔEN	Description	Energy	Example
Nonpolar covalent	≈ 0	Symmetric e^- sharing	150–500 kJ/mol	Cl_2, O_2
Polar covalent	$0 < \Delta EN \leq 1{,}7$	Asymmetric sharing, dipole	200–600 kJ/mol	HCl, H_2O
Ionic	$\Delta EN > 1{,}7$	e^- transfer, ions formed	400–1000 kJ/mol	NaCl, MgO
Metallic	≈ 0 (metals)	Delocalised e^- lattice	75–900 kJ/mol	Fe, Cu, Au

Covalent Bond

The *covalent bond* consists of the **sharing** of one or more electron pairs. Single bond = 1σ; double = $1\sigma + 1\pi$; triple = $1\sigma + 2\pi$.

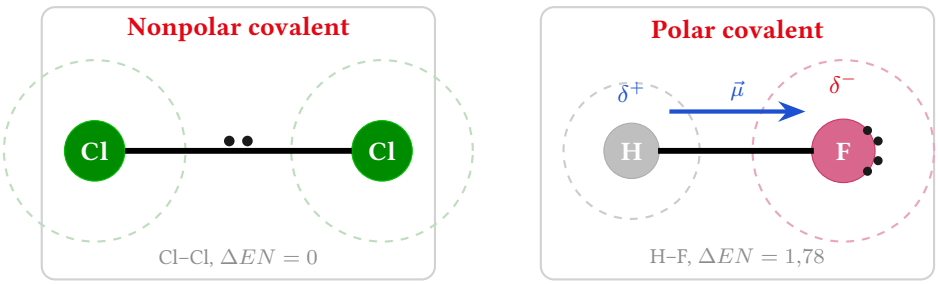

Figure 9: Nonpolar covalent bond (Cl_2) and polar covalent bond (HF).

Ionic Bond

Forms when $\Delta EN > 1{,}7$: the metal **donates** electrons (cation), the non-metal **accepts** them (anion). The lattice energy is:

$$U \propto \frac{|z^+| \cdot |z^-|}{r^+ + r^-}$$

NaCl: $\Delta EN = 3{,}16 - 0{,}93 = 2{,}23 \Rightarrow \mathrm{Na}^+$ [conf. Ne] + Cl^- [conf. Ar].

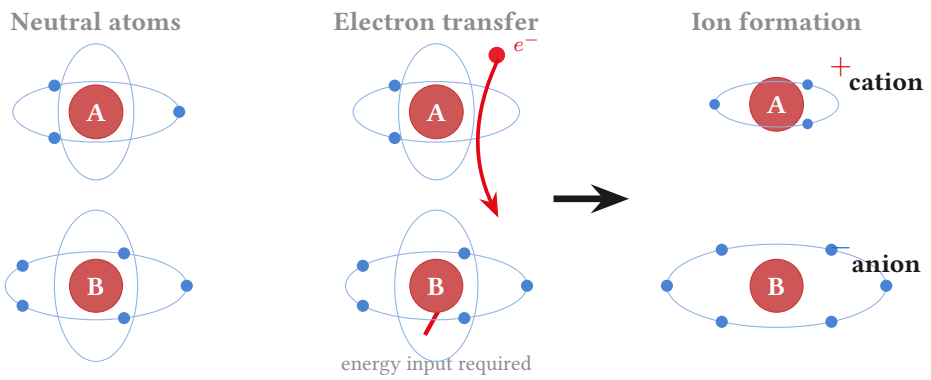

Figure 10: Ion formation: energy input triggers electron transfer from A to B. A loses an electron ⇒ **cation** (smaller, positive charge); B gains an electron ⇒ **anion** (larger, negative charge).

Metallic Bond

Metal cations are immersed in a *sea* of delocalised electrons (conduction band). This model explains electrical and thermal conductivity, ductility, malleability and lustre.

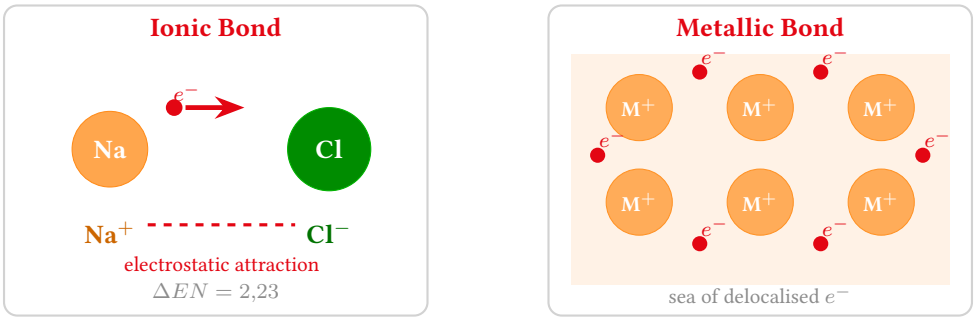

Figure 11: Ionic bond (NaCl): electron transfer and electrostatic attraction. Metallic bond: cations in a sea of free electrons.

6.2 Intermolecular Bonds

Intermolecular bonds act *between* distinct molecules and govern **macroscopic physical properties**: melting and boiling points, viscosity, surface tension. Stronger intermolecular forces raise the boiling point.

Type	Strength	Present in	Origin
London force (instant. dip. – induced dip.)	Weak ∼ 0,1–10 kJ/mol	**All** substances	Quantum fluctuations
Debye force (perm. dip. – induced dip.)	Moderately weak	Polar + polarisable mol.	Perm. dipole distorts cloud
Keesom force (perm. dip. – perm. dip.)	Moderate	Polar molecules	Alignment of opposite poles
Hydrogen bond X–H· · · Y	Strong 10–40 kJ/mol	H bonded to F, O or N	X–H$^{\delta+}$ · · · :Y

London Dispersion Forces

London forces arise from **instantaneous fluctuations** in electron density, creating a temporary dipole that induces a dipole in the neighbouring molecule. Strength is proportional to **polarisability**.

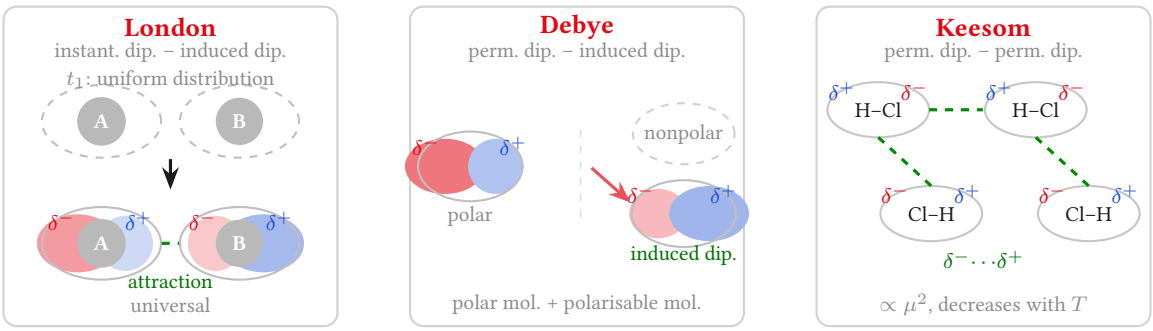

Figure 12: London, Debye and Keesom forces.

Debye Force

Acts between a polar molecule and a nonpolar molecule. The **permanent dipole** distorts the electron cloud of the neighbour, inducing a **temporary dipole**.

Keesom Force

Acts between molecules with **permanent dipoles**: they orient with opposite poles facing ($\delta^+ \cdots \delta^-$). Strength $\propto \mu^2$, **decreases with temperature**.

Hydrogen Bond

A **hydrogen bond** forms when an H atom covalently bonded to X approaches a second electronegative atom Y with a lone pair:

$$\text{X}{-}\text{H}^{\delta+} \cdots : Y \qquad \text{with X, Y} \in \{\text{F, O, N}\}$$

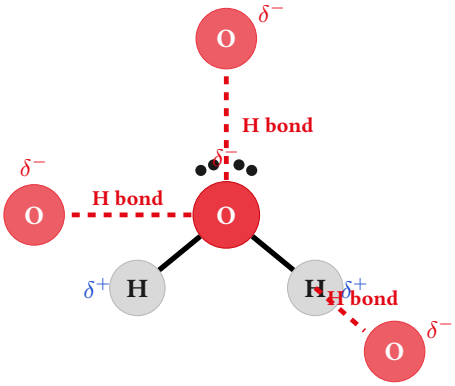

Figure 13: Hydrogen bond network in water. Each H_2O can form up to 4 hydrogen bonds (2 donor, 2 acceptor).

Anomalous boiling points. H_2O, HF and NH_3 have T_b significantly **higher** than analogues in their group due to hydrogen bonding. Ice is less dense than liquid water because the H-bond lattice is less compact.

Fundamental distinction. Breaking an **intramolecular** bond is a **chemical reaction** (e.g. O–H bond: ~ 460 kJ/mol). Breaking an **intermolecular** bond is a **physical process** (e.g. H bond in water: ~ 20 kJ/mol).

The feet of the *gecko* are covered by billions of very fine hairs. Each hair attaches via London forces ($\sim$nN per hair); collectively they support the animal's weight.

7 Valence Bond Theory — σ and π Bonds

Valence Bond theory (VB) describes the covalent bond as the overlap of two atomic orbitals, one from each atom. The overlap creates a region of high electron density between the two nuclei that lowers the system energy.

7.1 σ Bond (sigma)

The σ bond forms by *axial overlap* along the internuclear axis. It has **cylindrical symmetry**, allows **free rotation**, and is **always the first bond** to form.

7.2 π Bond (pi)

The π bond forms by *lateral overlap* of parallel p orbitals. It has a **nodal plane**, **prevents free rotation** (origin of *cis/trans* isomerism), is less stable than σ, and forms **only after** a σ bond exists.

$H_2C{=}CH_2$: C=C $= 1\,\sigma + 1\,\pi$; planar, 120.
HC≡CH: C≡C $= 1\,\sigma + 2\,\pi$; linear, 180.

Property	σ Bond	π Bond
Type of overlap	Axial (head-on)	Lateral (side-on)
Nodal plane on axis	No	Yes
Free rotation	Yes	No (cis/trans isomerism)
Relative stability	Greater	Lower
Formation order	First	Only after σ
Multiple bonds	Single = 1σ	Double = σ+π; triple = σ+2π

8 Orbital Hybridisation

Hybridisation is the linear combination of atomic orbitals of the *same atom* to form equivalent *hybrid orbitals* with the same energy and a larger lobe, optimised for directional σ bonds.

Hybrid orbitals describe **only** σ bonds and lone pairs. The π bonds form from pure p orbitals **not** involved in hybridisation.

8.1 sp^3 Hybridisation

$1\,s + 3\,p \;\Rightarrow\;$ 4 equivalent sp^3 hybrid orbitals directed towards a **tetrahedron** (109,5).

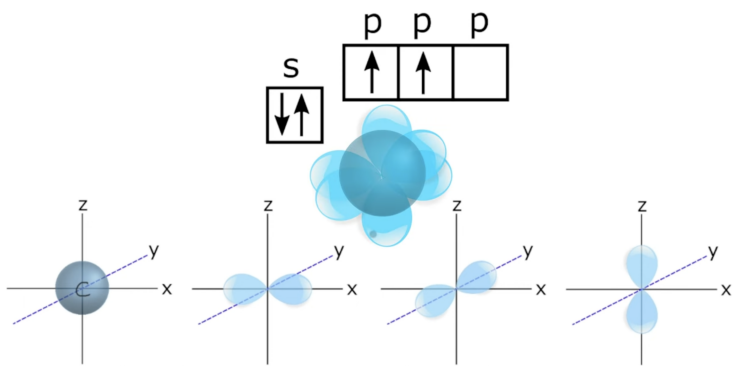

Figure 14: sp^3 hybridisation of carbon: the four equivalent hybrid orbitals orient tetrahedrally.

CH_4: tetrahedral, 109,5. **NH_3**: pyramidal, 107,3. **H_2O**: bent, 105.

8.2 sp^2 Hybridisation

$1\,s + 2\,p \;\Rightarrow\;$ 3 in-plane sp^2 orbitals (120) + 1 perpendicular p orbital $\Rightarrow$ 1 π bond possible.

$H_2C{=}CH_2$: planar, 120.

8.3 sp Hybridisation

$1\,s + 1\,p \;\Rightarrow\;$ 2 linear sp orbitals (180) + 2 pure p orbitals $\Rightarrow$ up to 2 π bonds.

$HC{\equiv}CH$: linear, 180.

8.4 Practical Rule: How to Determine Hybridisation

El. groups	Hybridisation	Electronic geometry	Free p orb.	π bonds
2	sp	Linear	2	up to 2
3	sp^2	Trigonal planar	1	1
4	sp^3	Tetrahedral	0	0
5	sp^3d	Trigonal bipyramid.	–	–
6	sp^3d^2	Octahedral	–	–

Multiple bond ⇒ reduced hybridisation. Double bond ⇒ sp^2; triple bond ⇒ sp.

Molecule	Hybrid.	Geometry	Angle	σ / π
CH_4	sp^3	Tetrahedral	109,5	4σ, 0π
NH_3	sp^3	Pyramidal	107,3	3σ, 0π
H_2O	sp^3	Bent	105	2σ, 0π
$H_2C{=}CH_2$	sp^2	Planar	120	5σ, 1π
$HC{\equiv}CH$	sp	Linear	180	3σ, 2π
CO_2	sp	Linear	180	2σ, 2π
BF_3	sp^2	Trigonal planar	120	3σ, 0π